

Dyadic-Comb Interpolation of the Fabius and Rvachev Functions

Runge instability, endpoint-jet constraints, exact arithmetic, and stable local alternatives

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Abstract

Let $N = 2^n$ and interpolate the Fabius function F at the additive dyadic comb $x_j = j/N$, or Rvachev's compactly supported function up at the affinely equivalent comb on $[-1, 1]$. This report separates three questions that are often conflated: accuracy for the particular Fabius data, conditioning of the value-data operator, and numerical evaluation of the resulting polynomial. Exact dyadic values are generated from the repository's Thue–Morse and moment formulas, so the observed oscillations are not caused by inaccurate samples.

For ordinary global Lagrange interpolation, high-precision experiments show a clear Runge-type boundary layer for both F and up: by $N = 64$ the sampled maximum errors are approximately 3.6×10^7 and 1.0×10^{10} , respectively. Imposing zero endpoint derivatives through order r can postpone the visible catastrophe, but it does not remove the underlying instability. The endpoint-flat Hermite interpolant has the exact factorization

$$H_{N,r}^F(x) = S_r(x) + [x(1-x)]^{r+1}Q_{N,r}(x),$$

where S_r is the beta-polynomial smoothstep and $Q_{N,r}$ is an ordinary Lagrange interpolant on the interior dyadic nodes. This gives explicit cardinal functions and permits sharp lower bounds. Every fixed r leaves exponential instability. More strongly, no sequence $r = r(N)$ can stabilize the value-data map: for all sufficiently large dyadic N , its Lebesgue constant is bounded below by Ce^{eN} uniformly for every $r \geq 0$. If first derivatives are instead prescribed at every node, a central value cardinal grows as $(2/\pi^2)4^N/N^4$, so global confluent interpolation can be even worse.

The function-specific error nevertheless has a narrow U-shaped window. Balancing two explicit cardinal modes predicts

$$r_{\text{bal}}(N) = \frac{2N \log 2}{2 \log N - 3 \log 2} = \frac{2N}{2 \log_2 N - 3} \quad (N = 2^n),$$

which matches the observed favorable orders (8, 14, 23) for $N = (32, 64, 128)$ to within one. This balance is an accuracy heuristic, not a stability theorem; at $N = 128$ the optimized errors already deteriorate. Finally, cellwise Hermite interpolation on the same dyadic comb is proved uniformly stable and satisfies explicit algebraic error bounds derived from the exact derivative norms of F and up. Conjectures connect the optimal-jet window, Thue–Morse midpoint surpluses, and the repository's lower-Lambert endpoint phase.

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1 Executive conclusions and proof status

The word “Runge” can mean at least three different things in this setting. The global interpolation operator can be exponentially ill-conditioned; a particular exact data set can develop visible endpoint oscillations; and a floating-point implementation can fail to evaluate an otherwise well-defined polynomial. The report keeps these levels separate.

Question	Conclusion	Status
Are dyadic samples exact?	Yes. All input samples used here are rational values generated from a finite Thue–Morse/moment formula.	Repository theorem plus exact code.
Does ordinary global interpolation exhibit Runge behavior?	Yes numerically for both F and up; operator instability is rigorous.	Numerical for the specific functions; proved for the operator.
Do zero endpoint derivatives help?	They suppress the first boundary-jet defects and create a striking finite- N accuracy window.	Exact factorization plus numerical evidence.
Do finitely many endpoint derivatives cure the instability?	No. Every fixed derivative order leaves an exponential cardinal mode.	Proved here.
Can one choose $r = r(N)$ to stabilize the global map?	No. Two competing explicit cardinal modes give an exponential lower bound uniformly in all $r \geq 0$.	Proved here.
Does matching derivatives at every node help?	Not in the global confluent polynomial. Even first-order Hermite value cardinals grow at least like $4^N/N^4$.	Proved here.
What does work on the same comb?	Local Hermite or spline interpolation, or oversampled regularized polynomial approximation.	Local error theorem proved; other methods cited.
What is the most promising new phenomenon?	The favorable endpoint order is accurately predicted by $r_{\text{bal}} = 2N/(2 \log_2 N - 3)$, despite unavoidable operator instability.	Derived heuristic and conjecture.

Scope and priority disclaimer

The theorems marked “proved here” have complete paper proofs in this report and were checked symbolically and numerically, but they have not yet been formalized in Lean or independently peer reviewed. The novelty audit establishes nonduplication with the specified repository and a targeted interpolation literature search; it is not a claim that no mathematically equivalent statement exists anywhere in the full historical literature. Conjectures and sampled-error claims are labeled explicitly.

2 Repository and literature audit

2.1 How the TeX corpus was read

The repository snapshot contains a current human-readable synthesis, a separate non-formalized frontier volume, a glossary and Lean walkthrough, specialized papers, and a frozen historical archive. The archive README explains that byte-identical duplicates were pruned and that the historical sources are retained for provenance rather than as competing current expositions. The audit therefore used the following route.

1. The current primary exposition `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex` was read for the canonical definitions, exact dyadic formulas, derivative laws, flatness, moments, q -binomial material, Fourier products, Lambert asymptotics, and the current formal-status map.

2. The current non-formalized-research-frontiers/non-formalized-research-frontiers.tex was searched and read around every occurrence of Lagrange, polynomial approximation, dyadic values, Richardson extrapolation, and geometric-node filters.
3. Every archived TeX title and source lineage listed in archive/README.md was cross-checked. Full-text searches were made for Runge, Hermite, Lebesgue, interpolation, and combinations of dyadic with Lagrange. Historical passages on Legendre expansion and best polynomial approximation were inspected separately.
4. The paper-source directories were enumerated and checked for the same interpolation vocabulary, with special attention to the Arias de Reyna sources and the lacunary-product material.

This process found substantial neighboring material but no treatment of global additive comb interpolation of F or up , no Runge analysis for those data, and no endpoint-flat Hermite construction of the form studied below.

2.2 What is already in the repository

Several existing components are essential inputs rather than new results of this report.

- Exact arbitrary-dyadic evaluation is available through finite Thue–Morse sums and moment recurrences. The values are rational, invariant under dyadic refinement, and implemented in Lean.
- The sharp derivative laws and endpoint flatness are proved. In the normalization used here,

$$\|F^{(m)}\|_{\infty,[0,1]} = \|\text{up}^{(m)}\|_{\infty,[-1,1]} = 2^{\binom{m+1}{2}} \quad (m \geq 0).$$

- The current primary volume develops a Fourier–Legendre expansion and least-squares interpretation. Orthogonal projection is not equispaced interpolation and does not have the same Runge mechanism.
- The repository’s Lagrange and Richardson theory concerns geometric nodes, including q -binomial weights and denominator-free filters. Those nodes are multiplicative, such as -2^k or q^k , whereas the present nodes are the additive comb $j/2^n$.
- The lower-Lambert endpoint asymptotics include a nonconstant periodic correction. They are potentially relevant to the location of interpolation boundary layers, but the current corpus does not connect them to Lagrange or Hermite interpolation.

2.3 Classical interpolation context

Runge’s original example showed that increasing the degree on equally spaced nodes can make an interpolant diverge near the endpoints [9]. The mechanism is summarized by the Lebesgue constant. For $N + 1$ equidistant nodes,

$$\Lambda_N^{\text{eq}} \sim \frac{2^{N+1}}{eN \log N}, \quad (1)$$

see Mills and Smith [10]. Barycentric evaluation is the standard stable way to evaluate the polynomial once the interpolation problem has been posed [11]; it does not change the condition number of the map from data to polynomial. More generally, stable recovery from equispaced samples cannot simultaneously have unrestricted exponential convergence, a barrier made precise by Platte, Trefethen, and Kuijlaars [12]. Oversampled polynomial frames provide one constructive way around the interpolation barrier [13].

The present functions are unusual test cases. They are C^∞ and compactly supported or bounded, but nowhere analytic in their nontrivial region. They are also infinitely flat at the relevant endpoints. Thus a local Taylor series predicts zero while the function is not zero, and a global polynomial is asked to reproduce a very thin nonanalytic transition using equispaced information. This combination makes derivative constraints both tempting and dangerous.

3 Fabius and Rvachev data on a dyadic comb

3.1 Normalization and structural identities

Let $F : [0, 1] \rightarrow [0, 1]$ denote the bounded Fabius function normalized by

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x). \quad (2)$$

Rvachev's even compactly supported function is

$$\text{up}(t) = \begin{cases} F(1 - |t|), & |t| \leq 1, \\ 0, & |t| \geq 1. \end{cases} \quad (3)$$

On $[0, 1]$ one has

$$F'(x) = 2 \text{up}(2x - 1). \quad (4)$$

Both functions are infinitely flat at their outer endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0, \quad \text{up}^{(m)}(-1) = \text{up}^{(m)}(1) = 0 \quad (m \geq 1). \quad (5)$$

The exact derivative growth is

$$\|F^{(m)}\|_\infty = \|\text{up}^{(m)}\|_\infty = 2^{m(m+1)/2}. \quad (6)$$

These facts, including attainment of the bound, are part of the current formal development [1].

For a dyadic level n put

$$N = 2^n, \quad x_j = \frac{j}{N}, \quad 0 \leq j \leq N. \quad (7)$$

The up comb on $[-1, 1]$ is $t_j = -1 + 2j/N$ and is carried to the same x_j by $t = 2x - 1$. This affine equivalence will allow the conditioning analysis to be stated once.

3.2 Exact Thue–Morse evaluation

Let $\varepsilon_h = (-1)^{\text{wt}_2(h)}$ and define the rational half moments d_j by

$$d_0 = 1, \quad d_m = \frac{1}{2^m - 1} \sum_{q=1}^m \binom{m}{q} \frac{d_{m-q}}{q+1} \quad (m \geq 1). \quad (8)$$

An equivalent form of the repository's closed dyadic formula is

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^n \binom{n}{j} d_j (a - 1 - h)^{n-j} \quad (9)$$

for $0 \leq a \leq 2^n$, with $0^0 = 1$ and an empty outer sum equal to zero. Formula (9) is obtained from the even-moment form in the repository by the identity

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2z + 1)^{n-2k} = 2^n \sum_{j=0}^n \binom{n}{j} d_j z^{n-j}.$$

It is particularly convenient for streaming the complete comb.

Algorithm 3.1 (Streaming exact comb). Define

$$S_k(a) = \sum_{h=0}^{a-1} \varepsilon_h (a-1-h)^k.$$

Starting with $S_k(0) = 0$, update

$$S_0(a+1) = S_0(a) + \varepsilon_a, \quad (10)$$

$$S_k(a+1) = \sum_{q=0}^k \binom{k}{q} S_q(a), \quad k \geq 1. \quad (11)$$

Then compute

$$F(a/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{j=0}^n \binom{n}{j} d_j S_{n-j}(a).$$

The complete level- n grid requires $\mathcal{O}(2^n n^2)$ integer/rational operations and no floating-point arithmetic.

The supplied Python program implements [theorem 3.1](#) with `fractions.Fraction`. Exact checks include reflection $F(a/N) + F((N-a)/N) = 1$ and the values $F(1/2) = 1/2$, $F(1/4) = 5/72$.

4 Ordinary global Lagrange interpolation

Let P_N^F be the polynomial of degree at most N with

$$P_N^F(x_j) = F(x_j), \quad 0 \leq j \leq N. \quad (12)$$

Let P_N^{up} denote the corresponding interpolant to $\text{up}(t_j)$ on $[-1, 1]$.

4.1 Exact symmetry and compressed forms

Theorem 4.1 (Symmetry compression). *For even N ,*

$$P_N^F(1-x) = 1 - P_N^F(x), \quad (13)$$

$$P_N^{\text{up}}(-t) = P_N^{\text{up}}(t). \quad (14)$$

Consequently there exist rational-coefficient polynomials R_N and T_N such that

$$P_N^F(x) = x + x(1-x)(2x-1)R_N(x(1-x)), \quad \deg R_N \leq \frac{N}{2} - 2, \quad (15)$$

$$P_N^{\text{up}}(t) = (1-t^2)T_N(t^2), \quad \deg T_N \leq \frac{N}{2} - 1. \quad (16)$$

Proof. The polynomial $1 - P_N^F(1-x)$ takes the value $F(x_j)$ at every node because of (2); uniqueness gives (13). Hence $P_N^F(x) - x$ is antisymmetric about $1/2$ and vanishes at $0, 1$, and $1/2$. After division by $x(1-x)(2x-1)$ the quotient is reflection-symmetric, and every polynomial symmetric under $x \mapsto 1-x$ is a polynomial in $x(1-x)$. Rationality follows from rational data and rational nodes. The up statement is identical after using evenness and the zeros at $t = \pm 1$. $\square$

This halves the effective dimension and provides exact symmetry checks for code. It does not alter the conditioning: the compressed coefficients can still be enormous.

4.2 Newton form and endpoint-jet defects

Write $f_j = f(j/N)$ and Δ for the forward difference in j . On an equispaced grid,

$$P_N f(x) = \sum_{k=0}^N \binom{Nx}{k} \Delta^k f_0. \quad (17)$$

Expanding the falling factorial with the signed Stirling numbers $s(k, m)$ gives an exact formula for the spurious endpoint jet:

Proposition 4.2 (Endpoint-jet defect). *For $0 \leq m \leq N$,*

$$(P_N f)^{(m)}(0) = m! N^m \sum_{k=m}^N \frac{s(k, m)}{k!} \Delta^k f_0. \quad (18)$$

In particular,

$$(P_N f)'(0) = N \sum_{j=1}^N (-1)^{j-1} \binom{N}{j} \frac{f_j}{j}. \quad (19)$$

For Fabius data every exact derivative on the left should be zero, so (18) measures a purely interpolatory boundary defect.

Proof. Use $(Nx)_k = \sum_m s(k, m) N^m x^m$ in (17). Equation (19) also follows directly by differentiating the cardinal basis at zero. $\square$

Fixing the first r endpoint derivatives removes the first r defects in (18). This explains why low-order constraints can improve a finite grid. It does not show that the remaining global modes are controlled.

4.3 Classical Lebesgue amplification

For the full-node cardinal polynomials ℓ_j ,

$$\Lambda_N = \max_{0 \leq x \leq 1} \sum_{j=0}^N |\ell_j(x)| \quad (20)$$

controls data perturbations and appears in Lebesgue's lemma

$$\|f - P_N f\|_\infty \leq (1 + \Lambda_N) \inf_{p \in \Pi_N} \|f - p\|_\infty. \quad (21)$$

The best polynomial error tends to zero for continuous f , but the exponential factor (1) can overwhelm it. Since F and up are nowhere analytic, there is no Bernstein-ellipse exponential decay available to compensate for the equispaced Lebesgue constant.

5 Endpoint-flat Hermite interpolation

5.1 Definition

For $r \geq 0$, define $H_{N,r}^F \in \Pi_{N+2r}$ by

$$H_{N,r}^F(x_j) = F(x_j), \quad 0 \leq j \leq N, \quad (22)$$

$$(H_{N,r}^F)^{(k)}(0) = (H_{N,r}^F)^{(k)}(1) = 0, \quad 1 \leq k \leq r. \quad (23)$$

There are $N + 1 + 2r$ confluent conditions and $N + 2r + 1$ coefficients, so the interpolant is unique. The convention $r = 0$ gives ordinary Lagrange interpolation. Define $H_{N,r}^{\text{up}}$ analogously on $[-1, 1]$, with zero derivatives through order r at both outer endpoints.

5.2 The smoothstep factorization

Put

$$S_r(x) = \frac{\int_0^x t^r(1-t)^r dt}{B(r+1, r+1)} = \frac{(2r+1)!}{(r!)^2} \sum_{k=0}^r (-1)^k \binom{r}{k} \frac{x^{r+k+1}}{r+k+1}. \quad (24)$$

Then $S_r(0) = 0$, $S_r(1) = 1$, all derivatives of orders $1, \dots, r$ vanish at the two endpoints, and

$$S_r(1-x) = 1 - S_r(x). \quad (25)$$

Let

$$D_r(x) = [x(1-x)]^{r+1}. \quad (26)$$

Theorem 5.1 (Exact endpoint-flat factorization). *For every dyadic $N \geq 2$ and $r \geq 0$,*

$$\boxed{H_{N,r}^F(x) = S_r(x) + D_r(x)Q_{N,r}(x)}, \quad (27)$$

where $Q_{N,r} \in \Pi_{N-2}$ is the ordinary Lagrange interpolant at the interior nodes $x_j = j/N$, $1 \leq j \leq N-1$, of

$$q_j = \frac{F(x_j) - S_r(x_j)}{D_r(x_j)}. \quad (28)$$

Moreover,

$$Q_{N,r}(1-x) = -Q_{N,r}(x), \quad (29)$$

and therefore

$$H_{N,r}^F(x) = S_r(x) + D_r(x)(2x-1)R_{N,r}(x(1-x)), \quad \deg R_{N,r} \leq \frac{N}{2} - 2. \quad (30)$$

All coefficients are rational.

Proof. The difference $H_{N,r}^F - S_r$ has a zero of multiplicity at least $r+1$ at both endpoints, so it is divisible by D_r . The quotient has degree at most $(N+2r) - (2r+2) = N-2$ and is fixed uniquely by its values at the $N-1$ interior nodes, giving (28). Reflection of the defining conditions and uniqueness give $H_{N,r}^F(1-x) = 1 - H_{N,r}^F(x)$. Combining this with (25) and the symmetry of D_r yields (29). Division by $2x-1$ produces a reflection-symmetric polynomial, hence a polynomial in $x(1-x)$. Rationality follows from (9) and the rational coefficients of S_r . $\square$

Corollary 5.2 (Rvachev factorization). *There is a polynomial $T_{N,r} \in \Pi_{N-2}$ such that*

$$H_{N,r}^{\text{up}}(t) = (1-t^2)^{r+1}T_{N,r}(t), \quad (31)$$

where $T_{N,r}$ interpolates $\text{up}(t_j)/(1-t_j^2)^{r+1}$ at the interior nodes. It is even and can be written as a polynomial in t^2 of degree at most $N/2 - 1$.

5.3 Explicit barycentric formula

For the interior nodes $x_j = j/N$, scaled barycentric weights are

$$w_j = (-1)^{j-1} \binom{N-2}{j-1}, \quad 1 \leq j \leq N-1. \quad (32)$$

Thus, away from the nodes,

$$Q_{N,r}(x) = \frac{\sum_{j=1}^{N-1} \frac{w_j q_j}{x - x_j}}{\sum_{j=1}^{N-1} \frac{w_j}{x - x_j}}. \quad (33)$$

This is the evaluation formula used in the experiments. High precision is essential: the barycentric formula prevents avoidable coefficient conversion errors, but it cannot remove the exponentially ill-conditioned dependence on q_j .

6 Why endpoint derivatives postpone but do not cure Runge instability

6.1 The value-data cardinal functions

Hold the endpoint values and endpoint derivatives fixed, and perturb one interior value. Let ℓ_j° be the ordinary cardinal polynomial for the interior grid $x_1, \dots, x_{N-1}$. The factorization immediately gives the value cardinal

$$L_{j,r}(x) = \left[\frac{x(1-x)}{x_j(1-x_j)} \right]^{r+1} \ell_j^\circ(x). \quad (34)$$

Define the corresponding Lebesgue function and constant by

$$\lambda_{N,r}(x) = \sum_{j=1}^{N-1} |L_{j,r}(x)|, \quad \Lambda_{N,r} = \max_{0 \leq x \leq 1} \lambda_{N,r}(x). \quad (35)$$

After $t = 2x - 1$, the ratio in (34) is unchanged. Therefore the value-data Lebesgue constants for F and u are identical.

6.2 Every fixed endpoint order remains exponentially unstable

Theorem 6.1 (Fixed-order boundary cardinal). *Let N be even, $j = N/2$, and evaluate at $x = 1/(2N)$. Then*

$$|L_{N/2,r}(1/(2N))| = \left[\frac{2}{N} \left(1 - \frac{1}{2N} \right) \right]^{r+1} \frac{\Gamma(N - \frac{1}{2})}{\sqrt{\pi}(\frac{N}{2} - \frac{1}{2})\Gamma(N/2)^2}. \quad (36)$$

For every fixed r ,

$$|L_{N/2,r}(1/(2N))| \sim \frac{2^{r+1}}{\sqrt{2}\pi} \frac{2^N}{N^{r+2}}. \quad (37)$$

In particular, $\Lambda_{N,r}$ grows exponentially with N for every fixed r .

Proof. At $Nx = 1/2$,

$$|\ell_{N/2}^\circ(x)| = \frac{\prod_{k=1, k \neq N/2}^{N-1} |\frac{1}{2} - k|}{(N/2 - 1)!^2} = \frac{\Gamma(N - \frac{1}{2})}{\sqrt{\pi}(\frac{N}{2} - \frac{1}{2})\Gamma(N/2)^2}.$$

The prefactor in (34) is the first factor in (36). Stirling's formula gives $|\ell_{N/2}^\circ(1/(2N))| \sim 2^N/(\sqrt{2}\pi N)$, proving (37). $\square$

The theorem makes precise what a low-order endpoint constraint does: each extra derivative contributes only an additional algebraic factor N^{-1} to this boundary mode. No fixed number of derivatives can defeat 2^N .

6.3 No varying endpoint order can stabilize the operator

One might let r grow with N . A single boundary mode can then be suppressed, but another mode becomes enormous. Two exact cardinal evaluations capture the competition.

For the central interior node $j = N/2$ at $x = 1/3$, dyadic N is never divisible by 3, and

$$A_{N,r} := |L_{N/2,r}(1/3)| = \left(\frac{8}{9} \right)^{r+1} \frac{3\sqrt{3}}{\pi N} \frac{\Gamma(N/3)\Gamma(2N/3)}{\Gamma(N/2)^2}. \quad (38)$$

At the first interior node $j = 1$ and the near-central point $x = 1/2 + 1/(2N)$,

$$B_{N,r} := |L_{1,r}(1/2 + 1/(2N))| = \left(\frac{N+1}{4}\right)^{r+1} \frac{\Gamma(N/2 - \frac{1}{2})^2}{\pi \Gamma(N-1)}. \quad (39)$$

Let

$$\delta = \log 2 - H(1/3) = \frac{5}{3} \log 2 - \log 3 = 0.056633012 \dots, \quad (40)$$

where $H(p) = -p \log p - (1-p) \log(1-p)$. Stirling's formula yields

$$A_{N,r} = \Theta\left(N^{-1} e^{\delta N} (8/9)^r\right), \quad (41)$$

$$B_{N,r} \sim \left(\frac{N+1}{4}\right)^{r+1} 2^{2-N} \sqrt{\frac{2}{\pi N}}. \quad (42)$$

Theorem 6.2 (Uniform all-order instability). *There exist absolute constants $c, C > 0$ and N_0 such that, for every dyadic $N \geq N_0$ and every integer $r \geq 0$,*

$$\boxed{\Lambda_{N,r} \geq C e^{cN}}. \quad (43)$$

Thus no choice $r = r(N)$ of endpoint derivative order stabilizes the global value-data interpolation operator.

Proof. Since a Lebesgue sum dominates every one of its terms, $\Lambda_{N,r} \geq \max(A_{N,r}, B_{N,r})$. Choose

$$\eta = \frac{\delta}{2 \log(9/8)}.$$

If $r \leq \eta N$, then (41) gives

$$\log A_{N,r} \geq \delta N - r \log(9/8) - \log N + \mathcal{O}(1) \geq \frac{\delta}{2} N - \log N + \mathcal{O}(1),$$

which is at least $c_1 N$ for large N . If $r > \eta N$, then (42) gives

$$\log B_{N,r} \geq \eta N \log\left(\frac{N+1}{4}\right) - N \log 2 + \mathcal{O}(\log N).$$

This is $\Omega(N \log N)$, hence exceeds $c_2 N$ for all sufficiently large N . Take $c = \min(c_1, c_2)$ and absorb finite prefactors into C . $\square$

Remark 6.3. The theorem is stronger than a fixed- r Runge statement. Raising r first damps the central-node boundary cardinal, but eventually the division by $[x_j(1-x_j)]^{r+1}$ makes data at nodes near the endpoint explosively influential in the interior. Endpoint flatness is therefore a preconditioner with two opposing regimes, not a stability cure.

6.4 Fixing derivatives at every node can be worse

A different reading of “fixing higher derivatives” is global confluent Hermite interpolation: prescribe values and derivatives at every grid node. Even the first-order case squares the dominant Lagrange cardinal.

Let $\mathcal{H}_N f$ be the degree- $2N+1$ polynomial matching f and f' at all $x_j = j/N$. If ℓ_j is the ordinary full-grid cardinal, the cardinal multiplying a perturbation of the value f_j is

$$h_j(x) = [1 - 2\ell'_j(x_j)(x - x_j)] \ell_j(x)^2. \quad (44)$$

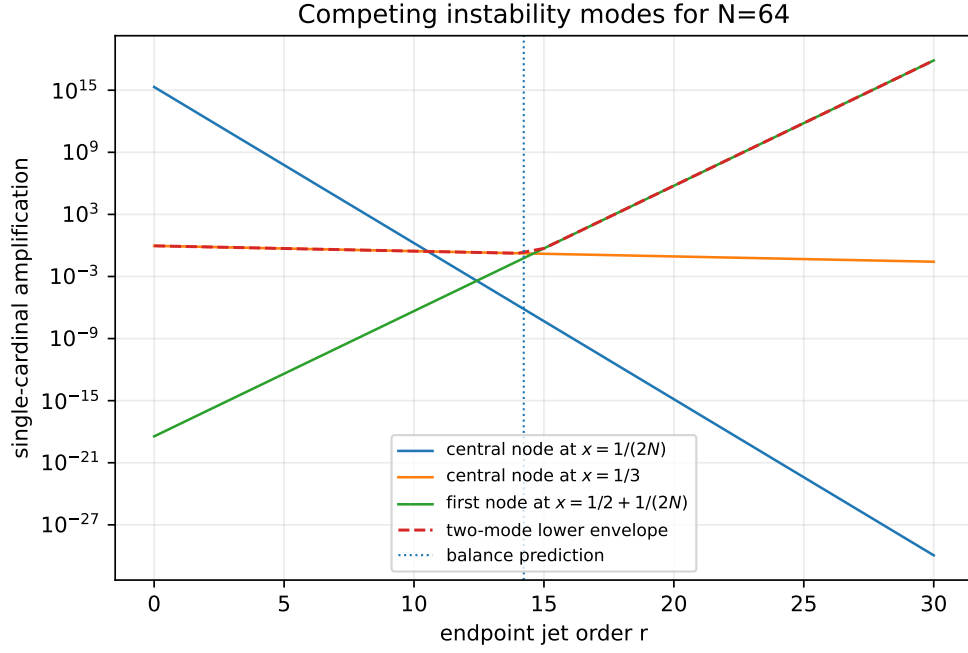


Figure 1: Three explicit value-cardinal modes for $N = 64$. Increasing r suppresses the half-cell and $x = 1/3$ modes but amplifies the first-interior-node mode. The dashed envelope used in [theorem 6.2](#) never becomes small.

Theorem 6.4 (First-order global Hermite amplification). *Let N be even and $j = N/2$. At $x = 1/(2N)$ one has $\ell'_j(x_j) = 0$ and*

$$|\ell_j(1/(2N))| = \frac{\Gamma(N + \frac{1}{2})}{2\sqrt{\pi}(\frac{N}{2} - \frac{1}{2})\Gamma(N/2 + 1)^2} \sim \frac{\sqrt{2}}{\pi} \frac{2^N}{N^2}. \quad (45)$$

Consequently the value-data part of the first-order global Hermite Lebesgue constant obeys

$$\Lambda_N^{C^1\text{-Hermite}} \geq |h_{N/2}(1/(2N))| \sim \frac{2}{\pi^2} \frac{4^N}{N^4}. \quad (46)$$

This lower bound applies even when all derivative data are exact and held fixed.

Proof. The derivative $\ell'_{N/2}(x_{N/2})$ is the symmetric sum $N \sum_{k \neq N/2} (N/2 - k)^{-1} = 0$. The gamma product in (45) follows by evaluating the full-node cardinal at $Nx = 1/2$. Equation (44) then reduces to ℓ_j^2 at this point, and Stirling's formula gives the asymptotic. $\square$

The conclusion is not that derivative information is useless. It is that forcing more confluent conditions into one global equispaced polynomial inherits, and can magnify, the same cardinal geometry. Local Hermite interpolation behaves differently because its cardinals never span an increasing number of cells.

7 A two-mode balancing law for the favorable jet window

The operator is unstable for every r , yet the exact Fabius and Rvachev data exhibit a narrow range in which large cardinal terms cancel unusually well. A useful prediction comes from balancing two boundary mechanisms rather than minimizing the rigorous envelope in [theorem 6.2](#).

Use the fixed- r half-cell mode from (37), whose leading logarithm is

$$N \log 2 + r(\log 2 - \log N), \quad (47)$$

and the near-central first-node mode (42), whose leading logarithm is

$$-N \log 2 + r(\log N - \log 4). \quad (48)$$

Equating (47) and (48) gives

$$r_{\text{bal}}(N) = \frac{2N \log 2}{2 \log N - 3 \log 2}. \quad (49)$$

For $N = 2^n$ this simplifies exactly to

$$r_{\text{bal}}(2^n) = \frac{2^{n+1}}{2n - 3} \sim \frac{N \log 2}{\log N}. \quad (50)$$

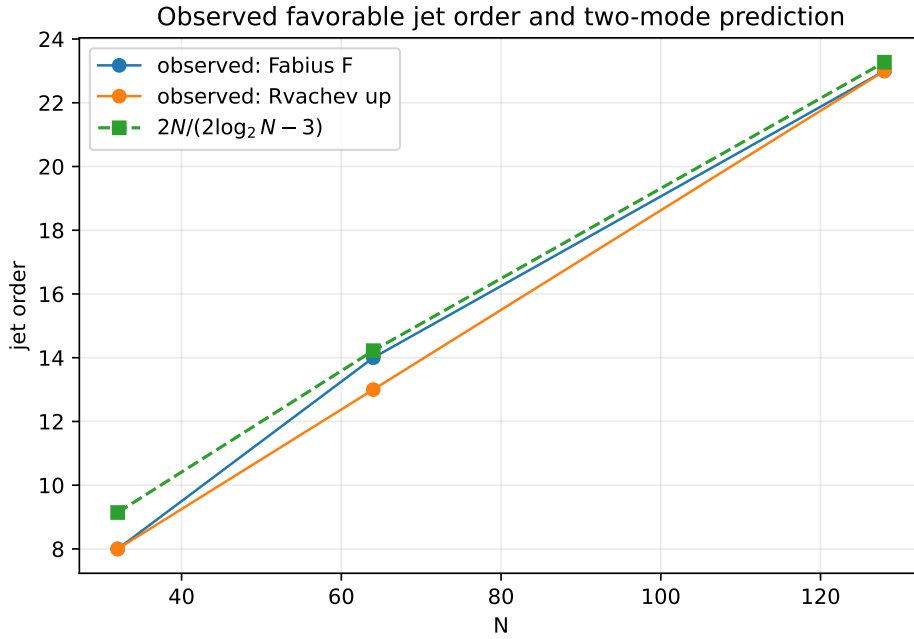


Figure 2: Observed minimizing orders in the scanned windows versus $r_{\text{bal}}(N) = 2N/(2 \log_2 N - 3)$. The agreement is much closer than a generic dimensional argument would suggest.

Target	N	observed r	r_{bal}	sampled best error
F	32	8	9.143	7.62×10^{-7}
up	32	8	9.143	4.08×10^{-5}
F	64	14	14.222	1.78×10^{-6}
up	64	13	14.222	2.12×10^{-4}
F	128	23	23.273	3.98×10^{-3}
up	128	23	23.273	5.31

Warning 7.1. The balance law predicts a favorable order for the particular data, not a small Lebesgue constant. At $r \asymp N/\log N$, the $x = 1/3$ cardinal from (38) is still exponentially large. The observed accuracy therefore relies on highly structured cancellation among exact Fabius values. Such cancellation is sensitive to perturbations.

8 Remainders, derivative growth, and dyadic hierarchy

8.1 The classical Hermite remainder is formally exact but practically weak

Proposition 8.1 (Endpoint-flat remainder). *If $f \in C^{N+2r+1}[0, 1]$, then for each $x \in [0, 1]$ there is $\xi_x \in (0, 1)$ such that*

$$f(x) - H_{N,r}f(x) = \frac{f^{(N+2r+1)}(\xi_x)}{(N+2r+1)!} x^{r+1}(x-1)^{r+1} \prod_{j=1}^{N-1} \left(x - \frac{j}{N}\right). \quad (51)$$

Proof. This is the standard confluent interpolation remainder for endpoint multiplicities $r+1$ and simple interior nodes. The nodal polynomial has total degree $2(r+1) + (N-1) = N+2r+1$. $\square$

For F or up, inserting (6) gives a factor $2^{(N+2r+1)(N+2r+2)/2} / (N+2r+1)!$. It is astronomically large and does not explain the small-error windows. This is not a contradiction: the derivative remainder uses one extreme global derivative norm and ignores cancellation. The failure is another analytic signature of nowhere analyticity.

A better theoretical route would combine the explicit cardinal functions with the exact Thue–Morse structure of the data, or use a Peano-kernel representation adapted to the self-similar probability law.

8.2 Nested dyadic correction identities

Dyadic grids are nested, so successive interpolants have exact factorizations.

Proposition 8.2 (Ordinary dyadic correction). *Let P_N and P_{2N} be ordinary interpolants to the same function on the two nested combs. Then*

$$P_{2N}(x) - P_N(x) = \omega_N(x) R_{N-1}(x), \quad \omega_N(x) = \prod_{j=0}^N \left(x - \frac{j}{N}\right), \quad (52)$$

where $R_{N-1} \in \Pi_{N-1}$ is determined by the N midpoint surpluses $f((2j+1)/(2N)) - P_N((2j+1)/(2N))$.

Proposition 8.3 (Endpoint-flat dyadic correction). *For fixed r ,*

$$H_{2N,r}(x) - H_{N,r}(x) = [x(1-x)]^{r+1} \prod_{j=1}^{N-1} \left(x - \frac{j}{N}\right) R_{N-1,r}(x), \quad (53)$$

with $R_{N-1,r} \in \Pi_{N-1}$ determined by the new midpoint data.

Proof. Both differences vanish at every coarse node. In the endpoint-flat case they also have zeros of multiplicity $r+1$ at 0 and 1. Degree counting leaves a quotient of degree at most $N-1$, and the N midpoint values determine it uniquely. $\square$

The factors in (52)–(53) define a natural hierarchical Runge diagnostic. If the normalized midpoint quotient becomes large near the boundary, the next global level is already signaling instability. Because the Fabius midpoint values are exact Thue–Morse rationals, this hierarchy can be studied without roundoff and may admit arithmetic recurrences.

9 High-precision numerical experiments

9.1 Method and reproducibility

The numerical pipeline has four safeguards.

1. Every interpolation datum is first computed as an exact rational by [theorem 3.1](#). The up data are obtained by the exact fold (3).
2. The endpoint-flat polynomial is evaluated by the interior barycentric formula (33). Working precision grows with N and r ; the largest scans used several hundred decimal digits.
3. Errors are compared with exact rational values on a finer dyadic grid. Tables labeled level K use $2^K + 1$ check points. They are sampled maximum errors, not certified continuous suprema.
4. Symmetry, interpolation, endpoint jets, and exact low-level values are checked independently. Increasing the check-grid level changes the quoted maxima only in the last reported digits for the displayed cases.

The archive contains `fabius_dyadic_interpolation_experiments.py`, all CSV data, and both PDF and PNG forms of the figures. The default mode is a manageable smoke test; `-mode report` performs the expensive high-precision scans.

9.2 Ordinary Runge growth and transient suppression

Target	N	$r = 0$	$r = 1$	$r = 2$	$r = 4$	$r = 8$
F	8	9.49×10^{-3}	1.75×10^{-3}	1.50×10^{-3}	1.64×10^{-3}	2.78×10^{-2}
F	16	3.80×10^{-3}	2.68×10^{-4}	1.37×10^{-4}	4.93×10^{-5}	1.44×10^{-4}
F	32	4.99	1.82×10^{-1}	9.75×10^{-3}	1.11×10^{-4}	7.62×10^{-7}
F	64	3.59×10^7	6.26×10^5	1.68×10^4	3.18×10^1	4.76×10^{-3}
up	8	2.02×10^{-1}	3.07×10^{-2}	1.96×10^{-2}	1.34×10^{-2}	2.55×10^{-1}
up	16	1.97×10^{-1}	2.67×10^{-2}	4.17×10^{-3}	9.85×10^{-4}	1.06×10^{-2}
up	32	4.50×10^2	1.46×10^1	6.76×10^{-1}	4.45×10^{-3}	4.08×10^{-5}
up	64	1.05×10^{10}	1.75×10^8	4.52×10^6	7.72×10^3	9.49×10^{-1}

Table 1: Sampled maximum errors on a level-11 dyadic check grid. Here $r = 0$ is ordinary Lagrange interpolation.

Ordinary interpolation looks innocuous at $N = 8$ and 16 but undergoes a dramatic transition between $N = 16$ and $N = 32$. The oscillation is concentrated in endpoint layers, as in the classical Runge phenomenon. Because the samples and comparison values are exact rationals, this is a property of the global interpolant, not sample noise.

9.3 The U-shaped jet scan

At $N = 64$, scanning all $0 \leq r \leq 20$ reveals the predicted tradeoff.

r	sampled error for F	sampled error for up
0	3.59×10^7	1.05×10^{10}
4	3.18×10^1	7.72×10^3
8	4.76×10^{-3}	9.49×10^{-1}
10	1.55×10^{-4}	2.73×10^{-2}
12	8.30×10^{-6}	1.26×10^{-3}
13	1.94×10^{-6}	2.12×10^{-4}
14	1.78×10^{-6}	6.19×10^{-4}
16	7.79×10^{-5}	1.95×10^{-2}
18	3.25×10^{-3}	7.31×10^{-1}
20	1.72×10^{-1}	3.65×10^1

Table 2: Selected rows of the $N = 64$ jet scan.

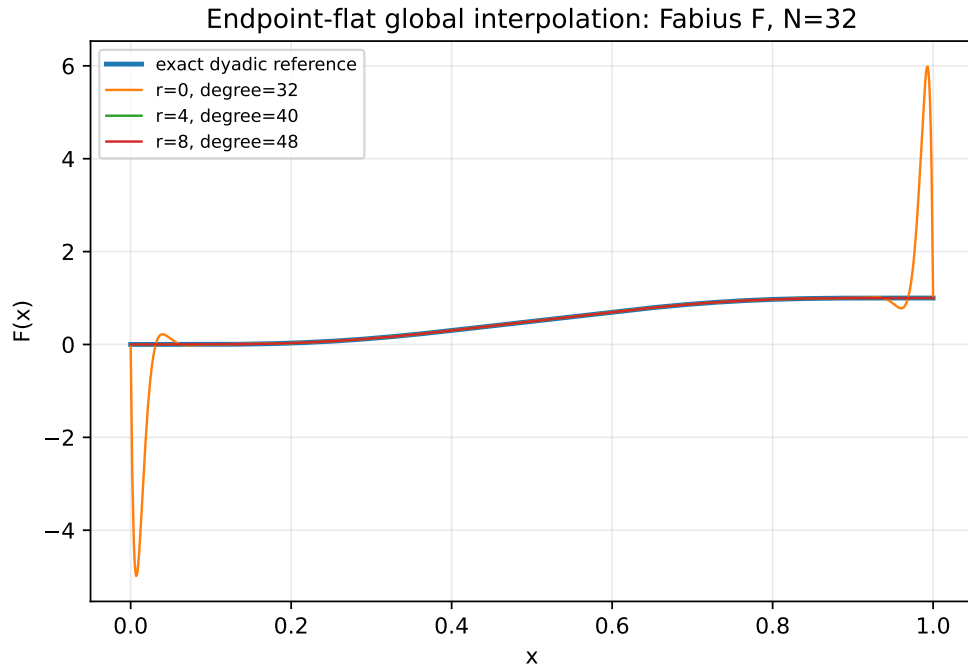


Figure 3: Fabius interpolation on the $N = 32$ comb. Ordinary interpolation has already entered a large boundary-layer regime. Moderate endpoint constraints recover excellent accuracy, but this improvement is temporary.

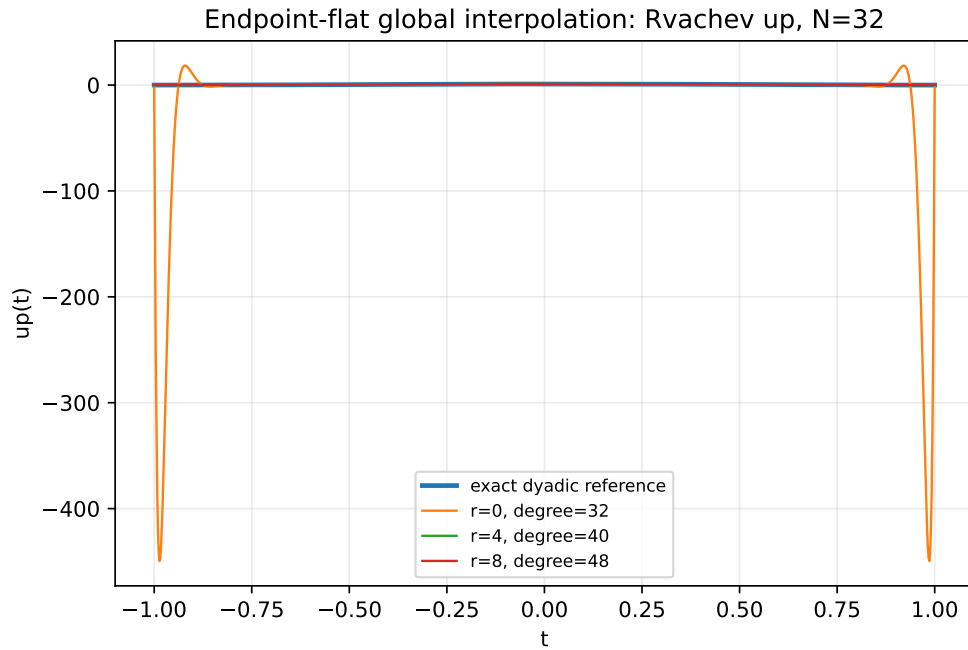


Figure 4: The affine $N = 32$ Rvachev experiment. Evenness is exact, so the two endpoint boundary layers are mirror images.

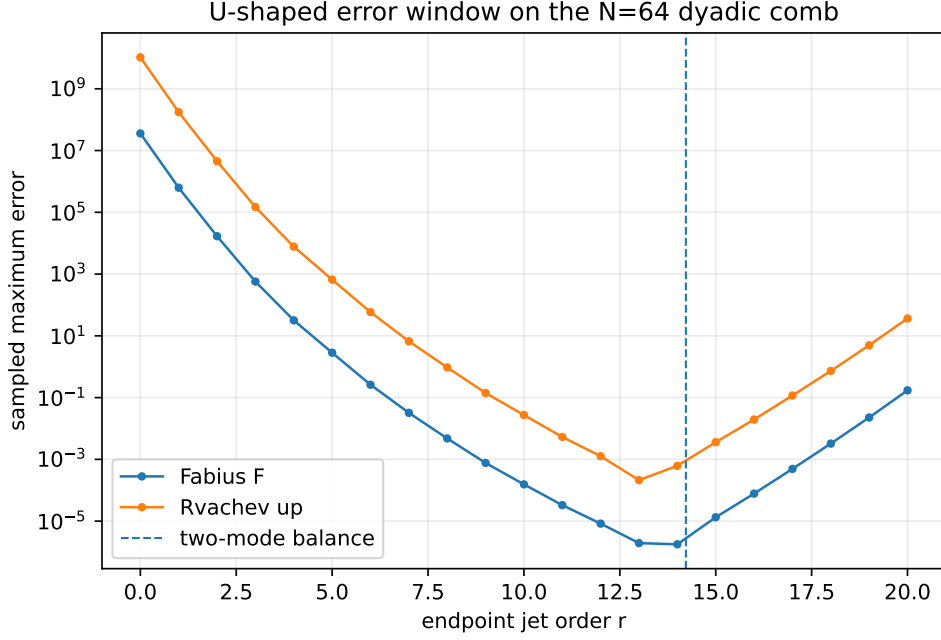


Figure 5: The error as a function of endpoint jet order is sharply U-shaped. The vertical line is $r_{\text{bal}}(64) = 128/9 = 14.22\dots$

The same phenomenon at $N = 128$ is less reassuring. The favorable order remains almost exactly the balancing prediction, but the minimum errors increase to about 3.98×10^{-3} for F and 5.31 for up. Thus optimizing r does not produce a monotone convergent method in the tested range.

9.4 Interpretation

The data support four conclusions.

1. *Runge phenomenon occurs for the specific functions.* The ordinary global interpolants develop rapidly growing endpoint oscillations.
2. *Correct endpoint derivatives matter at finite degree.* They remove the first spurious jets and can improve errors by many orders of magnitude.
3. *The benefit has a finite window.* Once r is too large, division by the tiny endpoint factors in (28) amplifies near-boundary data.
4. *Function-specific cancellation is not stability.* The small errors near the minimum coexist with exponentially large cardinal functions. Perturbing exact samples by even tiny relative errors can destroy the cancellation.

10 A stable cure on the same dyadic comb: local Hermite interpolation

The global polynomial is the source of the long-range cardinal coupling. Keep the same dyadic nodes but interpolate separately on each cell $[j/N, (j+1)/N]$.

For a fixed $s \geq 0$, let $\mathcal{H}_{N,s}^{\text{loc}} f$ be the piecewise polynomial of degree $2s+1$ that matches $f^{(k)}$ at both ends of every cell for $0 \leq k \leq s$.

Theorem 10.1 (Local endpoint-Hermite error). *For $f \in C^{2s+2}[0, 1]$,*

$$\|f - \mathcal{H}_{N,s}^{\text{loc}} f\|_{\infty} \leq \frac{\|f^{(2s+2)}\|_{\infty}}{(2s+2)!} \left(\frac{1}{4N^2} \right)^{s+1}. \quad (54)$$

Consequently,

$$\|F - \mathcal{H}_{N,s}^{\text{loc}} F\|_{\infty} \leq \frac{2^{(s+1)(2s+1)}}{(2s+2)!} N^{-2s-2}. \quad (55)$$

For N equal cells on $[-1, 1]$,

$$\|\text{up} - \mathcal{H}_{N,s}^{\text{loc}} \text{up}\|_{\infty} \leq \frac{2^{\binom{2s+3}{2}}}{(2s+2)!} N^{-2s-2}. \quad (56)$$

The local value and derivative cardinal functions have norms bounded independently of N .

Proof. On a cell $[a, b]$ of length h , the two-point Hermite remainder is

$$\frac{f^{(2s+2)}(\xi)}{(2s+2)!} (x-a)^{s+1} (x-b)^{s+1}.$$

The absolute product is at most $(h^2/4)^{s+1}$. For F , use $h = 1/N$ and (6); the factor $4^{-(s+1)}$ reduces the derivative exponent to $(s+1)(2s+1)$. For up on $[-1, 1]$, $h = 2/N$, so $h^2/4 = N^{-2}$ and no further power of four appears. After rescaling a cell to $[0, 1]$, the local cardinal basis depends only on s , proving N -independent conditioning. $\square$

For cubic Hermite interpolation ($s = 1$), exact dyadic values and exact derivatives from (4) give the following sampled errors.

N	4	8	16	32	64
sampled error	9.67×10^{-4}	5.03×10^{-4}	2.06×10^{-5}	2.29×10^{-6}	1.58×10^{-7}

Table 3: Piecewise cubic Fabius errors on a level-12 check grid.

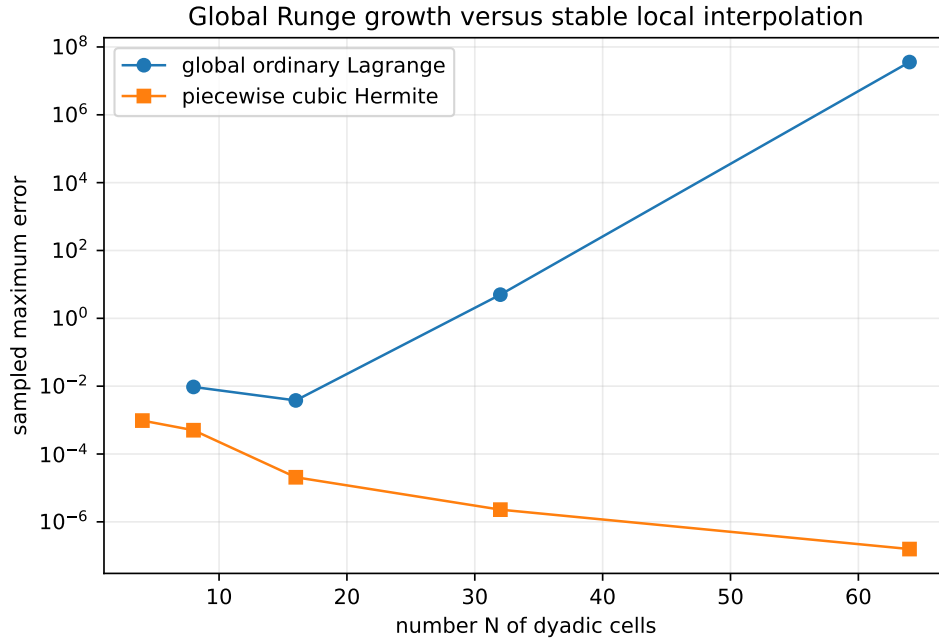


Figure 6: Ordinary global interpolation becomes catastrophic, while the cellwise cubic method converges steadily on the same dyadic samples.

Other stable choices that retain equispaced data include low-degree splines, Floater–Hormann rational interpolation, constrained least squares, and oversampled polynomial frames. If a single global polynomial is essential, oversampling and regularization are more promising than adding exact confluent conditions at every sample.

11 Connections with the surrounding Fabius–Rvachev theory

11.1 Thue–Morse signs and exact interpolation coefficients

Every coefficient of P_N^F and $H_{N,r}^F$ is rational and can be expanded into finite sums involving ε_h . Combining (17) with (9) gives a triple finite sum for the Newton coefficients $\Delta^k F(0)$. Prouhet cancellation annihilates low-degree polynomial parts of complete Thue–Morse blocks. This suggests that the unexpectedly small function-specific error near r_{bal} is not accidental: large cardinals may cancel against blockwise signed moment identities in the exact data.

A tractable first target is the midpoint surplus

$$\sigma_{N,j} = F\left(\frac{2j+1}{2N}\right) - P_N^F\left(\frac{2j+1}{2N}\right), \quad (57)$$

which is rational and determines the entire correction in (52). Its numerator should admit a block decomposition governed by the Thue–Morse polynomial $\prod_{k=0}^{n-1}(1 - z^{2^k})$.

11.2 Additive versus geometric Lagrange weights

The repository’s q -binomial interpolation uses geometric nodes and weights naturally expressed through Gaussian binomial coefficients. The additive interior comb instead has the binomial barycentric weights (32). Both are manifestations of a Vandermonde inverse, but with different node groups:

Geometry	Nodes	Natural weights
additive dyadic comb	$j/2^n$	$(-1)^{j-1} \binom{N-2}{j-1}$
geometric/ q comb	q^j or -2^j	Gaussian-binomial products

A mixed transform could interpolate across additive location while filtering across dyadic scale. The nested surplus polynomial in (53) is a natural place to search for such an additive–multiplicative bridge.

11.3 Fourier sinc products and local approximation

The infinite sinc product for $\widehat{\text{up}}$ explains the extraordinary smoothness and provides moment and derivative information. For local interpolation it yields useful Fourier-domain error estimates: the cellwise operator acts as a compactly supported quasi-interpolant whose multiplier can be compared with the sinc product near the origin. This may sharpen the elementary derivative bound (56), especially when s grows slowly with N .

By contrast, a global polynomial has no compactly localized Fourier response. Its endpoint oscillations correspond to broad spectral leakage, another way to see why the compact-support structure of up is better respected by local atomic-function or spline bases.

11.4 Lambert–W endpoint phase

The repository proves that the sharp small- x asymptotic of $F(x)$ contains a nonconstant periodic correction evaluated at a lower-Lambert phase. Global interpolation errors are largest in boundary layers whose locations shrink with N . Substituting the empirical maximizer x_N into the lower-Lambert phase may reveal a log-periodic drift in both the error and the integer offset $r_{\text{opt}} - r_{\text{bal}}$. This would connect a purely numerical-analysis phenomenon with the endpoint saddle structure.

11.5 Bell and Bernoulli polynomials

The moments c_k, d_k have Bernoulli recurrences, while saddle corrections are organized by Bell-polynomial algebra in the repository. Two possible uses in interpolation are:

1. express the exact endpoint jets (18) as Bell-polynomial transforms of the moment generating series;
2. derive asymptotic expansions of the gamma-cardinal modes beyond leading order, then balance them to obtain corrections to (49).

The second program is especially concrete. Stirling–Bernoulli expansions of (36) and (39) should produce

$$r_{\text{opt}}(N) = \frac{2N \log 2}{2 \log N - 3 \log 2} + \frac{a_1 N}{(\log N)^2} + a_0 + \cdots, \quad (58)$$

with coefficients modified by function-specific Thue–Morse cancellation.

11.6 Inverse Fabius interpolation

The inverse F^{-1} has infinite endpoint slopes rather than flat endpoints. Direct polynomial interpolation of F^{-1} on a uniform value comb should therefore have a different but equally severe endpoint problem. A more natural scheme is parametric: interpolate the exact pairs $(F(j/N), j/N)$ or apply a variable transformation informed by the lower-Lambert asymptotic. The endpoint-flat analysis here supplies a warning: imposing large derivative data globally is unlikely to be stable; local rational or monotone Hermite methods are preferable.

12 Conjectures

The following statements are deliberately separated from the proved results.

Conjecture 12.1 (Two-mode optimal-jet law). *Let r_N^F and r_N^{up} minimize the sampled or true uniform error among the endpoint-flat global interpolants. Along dyadic $N = 2^n$,*

$$r_N^F = \frac{2N}{2n-3} + \mathcal{O}(1), \quad r_N^{\text{up}} = \frac{2N}{2n-3} + \mathcal{O}(1). \quad (59)$$

A refined version has bounded oscillatory corrections determined by a Thue–Morse or lower-Lambert phase.

Conjecture 12.2 (Failure of optimized global convergence). *Define*

$$E_N^F = \inf_{r \geq 0} \|F - H_{N,r}^F\|_\infty, \quad E_N^{\text{up}} = \inf_{r \geq 0} \|\text{up} - H_{N,r}^{\text{up}}\|_\infty.$$

Then neither sequence tends to zero along all dyadic N . More strongly,

$$\limsup_{n \rightarrow \infty} E_{2^n}^F > 0, \quad \limsup_{n \rightarrow \infty} E_{2^n}^{\text{up}} = +\infty. \quad (60)$$

The $N = 128$ deterioration is the first numerical indication, not a proof.

Conjecture 12.3 (Dyadic surplus arithmetic). *After multiplication by the natural common denominator from the dyadic moment formulas, the midpoint surplus vector $\sigma_{N,j}$ in (57) factors into a finite binomial transform of a Thue–Morse block. Its sign changes and largest boundary entries obey a recursion of depth n .*

Conjecture 12.4 (Higher global confluent amplification). *For global Hermite interpolation matching derivatives through a fixed order $s \geq 1$ at all $N + 1$ equidistant nodes, the value-data Lebesgue constant has a lower bound of the form*

$$\Lambda_{N,s}^{\text{confluent}} \geq C_s \frac{2^{(s+1)N}}{N^{\alpha_s}} \quad (61)$$

for explicit α_s . Theorem 6.4 is the case $s = 1$ with $\alpha_1 = 4$.

Conjecture 12.5 (Lambert phase locking). *Let x_N be the leftmost point at which the optimized interpolation error reaches its maximum. The lower-Lambert phase associated with x_N has a limiting rotation modulo one, and the fluctuations of $r_N - r_{\text{bal}}(N)$ and $\log E_N$ are continuous periodic functions of that phase.*

13 Promising research directions

13.1 Prove function-specific divergence

The most important open step is to pass from operator instability to a lower bound for the actual Fabius data. Three routes look promising.

1. Insert the exact dyadic formula into a selected cardinal evaluation and exploit Thue–Morse block cancellation until one dominant term remains.
2. Analyze the Newton coefficients $\Delta^k F(0)$ by a two-variable saddle method in k and n .
3. Use the nested surplus factorization and prove that a boundary surplus cannot remain small at all dyadic levels.

13.2 Obtain a corrected balance expansion

Use Stirling–Bernoulli series for the two cardinal modes and match not only the exponential terms but the powers of N and constants. Compare the resulting correction with exact high-precision minimizers for $N = 256$ and 512 . The formula (58) should be tested separately for F and up.

13.3 Design stable dyadic approximants

The same exact samples can support methods with much better conditioning:

- piecewise Hermite interpolation of adaptive local order;
- compactly supported Rvachev atomic-function quasi-interpolation;
- monotone rational Hermite interpolation for F ;
- oversampled polynomial least squares with degree $m \ll N$;
- polynomial frames or constrained minimax reconstruction;
- rational barycentric methods with poles kept away from $[0, 1]$.

A useful benchmark suite should report approximation error, Lebesgue/condition number, monotonicity, symmetry defect, and exact-data sensitivity separately.

13.4 Exploit the dyadic hierarchy

The correction polynomials in (52) and (53) give an adaptive algorithm: compute midpoint surpluses exactly, stop refining cells whose normalized surplus is small, and switch from global to local interpolation when the boundary quotient begins to grow. This creates a mathematically transparent Runge detector specialized to nested dyadic data.

13.5 Formalize the new finite theorems

The factorization theorem, cardinal formulas, gamma-product identities, and all-order case split are finite algebra and real asymptotics well suited to Lean. A suggested sequence is:

1. confluent interpolation uniqueness and divisibility by endpoint powers;
2. interior barycentric weights on an arithmetic progression;
3. exact gamma products for the three selected cardinals;
4. Stirling-ratio asymptotics;

5. the uniform-in- r instability theorem;
6. local Hermite remainder and Fabius derivative specialization.

This would extend the repository’s current geometric Lagrange formalization to additive comb interpolation without duplicating it.

14 Summary

Global equispaced interpolation of the Fabius and Rvachev functions is a particularly sharp laboratory for Runge instability. Exact arithmetic removes sample uncertainty; symmetry reduces the polynomials to half-size forms; and endpoint flatness offers seemingly ideal Hermite data. Yet the geometry of the comb dominates. A fixed number of endpoint constraints changes exponential cardinal growth only by powers of N , while a growing number transfers the instability from central data to near-endpoint data. Matching first derivatives at every node can square the exponential growth.

The surprising positive result is not stability but structure. The exact smoothstep factorization exposes the mechanism, and a two-mode calculation predicts the narrow function-specific accuracy window almost perfectly. That window should be treated as a frontier cancellation phenomenon, probably tied to Thue–Morse arithmetic and endpoint Lambert phases, rather than as a reliable numerical method. On the same dyadic comb, local Hermite interpolation gives the genuine cure: bounded local conditioning and explicit algebraic convergence.

A Gamma-product calculations

A.1 Interior central cardinal at a half cell

For interior nodes $1, \dots, N-1$ in the scaled variable $y = Nx$ and $j = N/2$,

$$\ell_j^\circ(y/N) = \prod_{\substack{1 \leq k \leq N-1 \\ k \neq j}} \frac{y-k}{j-k}.$$

At $y = 1/2$,

$$\prod_{k=1}^{N-1} |k - 1/2| = \frac{\Gamma(N - 1/2)}{\Gamma(1/2)},$$

and deleting the j factor divides by $j - 1/2$. The denominator is $(j-1)!(N-1-j)! = \Gamma(N/2)^2$, yielding (36).

A.2 The one-third mode

At $y = N/3$,

$$\prod_{k=1}^{N-1} (N/3 - k) = \frac{\Gamma(N/3)}{\Gamma(1 - 2N/3)}.$$

Reflection gives

$$\frac{1}{|\Gamma(1 - 2N/3)|} = \frac{\Gamma(2N/3) |\sin(2\pi N/3)|}{\pi} = \frac{\sqrt{3}}{2\pi} \Gamma(2N/3),$$

because dyadic N is not divisible by 3. Removing the central factor $|N/3 - N/2| = N/6$ and dividing by $\Gamma(N/2)^2$ gives (38). The exponential part of the gamma ratio is

$$\exp \left\{ N \left[\frac{1}{3} \log \frac{1}{3} + \frac{2}{3} \log \frac{2}{3} - \log \frac{1}{2} \right] \right\} = e^{N(\log 2 - H(1/3))}.$$

A.3 The near-central boundary-node mode

For $j = 1$ and $y = (N + 1)/2$,

$$\left| \prod_{k=2}^{N-1} (y - k) \right| = \left[\frac{\Gamma(N/2 - 1/2)}{\Gamma(1/2)} \right]^2.$$

The denominator is $(N - 2)! = \Gamma(N - 1)$. Also

$$\frac{x(1 - x)}{x_1(1 - x_1)} = \frac{(1/4)(1 - N^{-2})}{N^{-1}(1 - N^{-1})} = \frac{N + 1}{4},$$

proving (39). The duplication formula gives

$$\frac{\Gamma(N/2 - 1/2)^2}{\pi \Gamma(N - 1)} = \frac{2^{2-N}}{\sqrt{\pi}} \frac{\Gamma((N - 1)/2)}{\Gamma(N/2)} \sim 2^{2-N} \sqrt{\frac{2}{\pi N}}.$$

B Core exact-arithmetic pseudocode

Listing 1: Streaming the exact dyadic Fabius comb.

```
def fabius_grid_exact(level):
    n = level
    N = 2**n
    d = exact_half_moments(n)          # Fractions
    S = [0] * (n + 1)                  # integer Thue--Morse prefix moments
    scale = Fraction(1, 2**(n*(n-1)//2) * factorial(n))
    values = []

    for a in range(N + 1):
        total = sum(comb(n, j) * d[j] * S[n-j]
                     for j in range(n + 1))
        values.append(scale * total)
        if a == N:
            break

    eps = (-1)**popcount(a)
    next_S = [0] * (n + 1)
    next_S[0] = S[0] + eps
    for k in range(1, n + 1):
        next_S[k] = sum(comb(k, q) * S[q]
                        for q in range(k + 1))

    S = next_S
    return values
```

The actual program adds caching, reflection checks, exact derivative data, high-precision barycentric interpolation, CSV output, figures, cardinal-mode formulas, and command-line options.

C Reproduction manifest

The ZIP archive contains:

Path	Purpose
fabius_dyadic_interpolation_repo	Complete LaTeX source.
rt.tex	

Path	Purpose
fabius_dyadic_interpolation_repo rt.pdf	Rendered report.
fabius_dyadic_interpolation_experiments.py	Exact rational and high-precision numerical experiments, with extensive comments.
data/fixed_order_errors.csv	Errors for $r \in \{0, 1, 2, 4, 8\}$ and $N \leq 64$.
data/jet_scan_N64.csv	Complete $0 \leq r \leq 20$ scan for both targets.
data/jet_scan_N128_window.csv	Focused high-precision scan around the predicted favorable order.
data/observed_optimal_orders.csv	Comparison with the balancing law.
data/piecewise_cubic_errors.csv	Stable local-Hermite comparison.
data/cardinal_modes.csv	Explicit theoretical cardinal values near r_{bal} .
figures/*.pdf, figures/*.png	Vector figures used in the report and raster copies for inspection.
README.md	Build and execution instructions.

To rebuild the numerical outputs:

```
python fabius_dyadic_interpolation_experiments.py \
    --mode report --check-level 11 --output-dir reproduced
```

The full scan is intentionally expensive because it uses exact input arithmetic and hundreds of decimal digits for the most ill-conditioned cases. The default quick mode provides a smaller verification run.

D Corpus coverage matrix

Corpus component	Relevant existing content	Nonduplication conclusion
Current primary exposition	Exact dyadic values; derivatives and flatness; moments; Fourier–Legendre least squares; geometric Lagrange and Richardson; lower-Lambert endpoint asymptotics.	No additive-comb Runge or endpoint-flat Hermite analysis.
Current frontier volume	Non-formalized asymptotics, q -filters, dyadic formulas, approximation ideas, and conjectures.	No theorem equivalent to theorems 5.1, 6.2 and 6.4 .
Historical primary and supplements	Earlier versions of function theory, exact arithmetic, and asymptotics.	Keyword and passage audit found no ordinary/Hermite dyadic-comb study.
Historical dyadic and q studies	Alternative exact formulas, Gaussian coefficients, convergence accelerators.	Node geometry and objectives differ from additive global interpolation.
Historical frontier consolidation	Legendre expansion and best polynomial approximation among many frontier topics.	Orthogonal approximation is distinct from interpolation and Runge conditioning.
Paper-source directories	Arias de Reyna, lacunary products, Fabius and arithmetic source papers.	Supply background identities; no matching endpoint-jet Runge result found.

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